Future-Proofing STEM Education

Integrating AI Literacy and Immersive Technologies

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An AI-powered physics learning system that combines personalised tutoring, practice and learning reports with an interactive desk-lamp companion.

A CONTEXT & CORE CHALLENGE

Personal support across the whole study journey.

Physics students need timely feedback and clear explanations when they get stuck. The challenge is to connect each question with their learning history.

STEM Agent links tutoring, practice and reports through a shared learner profile, using previous work and preferences to guide each study session.

B CURRENT PLATFORM CAPABILITIES

Explore. Ask. Practise. Review.

Explore physics & meet your tutor.

PHYSICS NEWS

SCIENCE TRIVIA

SEVEN TUTOR PERSONAS

01 / TUTORING

Bring a question to the tutor.

Follow a guided explanation, ask follow-up questions and reflect on understanding.

02 / PRACTICE

Work through a practice session.

Answer questions with a timer, track progress and star items to revisit.

03 / LEARNING REPORTS

Review the learning record.

Review recent activity, knowledge points and weekly summaries; export reports as PDF.

04 / PERSONAL SETTINGS

Personalise your tutor.

Set explanation style, choose a tutor persona and control optional emotion sensing.

05 / REWARDS & UNLOCKS

Unlock tutors & badges.

Unlock tutor cards and collect physics-themed badges as you study.

C INTERDISCIPLINARY INTEGRATION

Five contributions connected through one learner profile.

AI supports learner modelling and teaching; IoT connects the study space; media design shapes interaction.

SENSING
Vision & affect
Process camera input for optional facial emotion sensing.

MEMORY
Learner profile
Store learning history and preferences in a shared learner profile.

TEACHING
Teaching policy
Select hints, explanations and examples using the learner's state.

PRACTICE
Practice & reports
Configure practice, track progress and return learning reports.

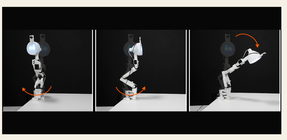
EMBODIMENT
Desk-lamp companion
Provide study cues through the lamp's movement and light.

LUMI / LIGHT, MOVEMENT & VOICE

VISUAL CUES

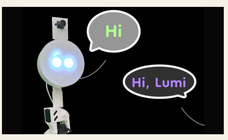
Lighting, symbols, text & expressions.

KINETIC CUES



Rotate, nod, extend and lean.

VERBAL CUES



Voice prompts and dialogue.

D IMPACT, LIMITATIONS & FUTURE DIRECTIONS

From a working platform to classroom evaluation.

INTENDED VALUE
Continuous learning support for students; clearer records of learning needs for instructors.

CURRENT LIMIT
Classroom learning benefits still require a controlled study.

NEXT STEP
Evaluate usability, reliability and learning retention in a supervised classroom pilot.

LIVE DEMO
139.199.17.201

VOTING QR